

Low frequency noise assessment in n- and p-channel sub-10 nm triple-gate FinFETs: Part I: Theory and methodology



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ABSTRACT

The transfer characteristic at room temperature of FinFETs processed for sub-10 nm technologies could always be explained by solving Poisson equation throughout the channel – dielectric interface. Various methods for the MOSFET parameters estimation are proposed in the literature. In this paper, the electrical parameters extraction technique based on the Y-function methodology is reminded.

Low frequency noise is presented considering three major noise sources: 1/f noise associated to carrier trapping-detrapping in the gate oxide, channel carrier mobility fluctuations and generation-recombination noise related to traps located in the depletion zone of the device. Theory and methodology in order to identify the 1/f noise mechanism and to have information of the process induced traps in the silicon film using the noise spectroscopy technique are revisited.

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1. Introduction

Low frequency noise measurements can be used as a non-destructive diagnostic tool that leads to the identification of traps in the Si and the gate oxide, thus giving informations on the quality of the transistors fabrication. The study of the 1/f noise level with respect to the gate overdrive voltage V_{GT} leads to the identification of the 1/f noise origin. The study of the generation-recombination noise, if performed as a function of the temperature at fixed drain currents, gives access to information on the traps that are located in the depletion region of the transistor [1].

This paper summarizes the extraction methods and techniques that have been used in order to investigate triple-gate FinFETs issued from a sub-10 nm technological node.

Based on the expression of the drain current which takes into account surface roughness scattering impact on the effective carrier mobility, simple and accurate method for parameter extraction is based on the $Y = I_D/\sqrt{g_m}$ function. Indeed, the Y function exhibits a linear variation on the applied gate voltage and does not depend on the access resistance through the first mobility attenuation factor. Special attention is given for the threshold voltage,

low field carrier mobility and access resistance estimation which are needfully for 1/f low frequency noise modeling.

Then the low frequency noise theory is discussed, 1/f noise characterisation is described as it leads to the identification of the 1/f noise mechanism and an estimation of the gate oxide trap density. Finally the study of the depletion region traps is explained using a low frequency noise spectroscopy as a function of temperature.

2. Static parameters extraction

2.1. MOSFET model in linear operation region

MOSFETs are composed from three resistive regions: the source, the channel and the drain, as seen in Fig. 1. In the linear region of operation, the MOSFET drain current I_D in strong inversion can be expressed by [2]

$$I_D = \frac{W}{L} \mu_{eff} |Q_i| V_{D'S'} \quad (1)$$

W and L are the gate effective width and length, respectively, $V_{D'S'}$ is the voltage across the channel (between the points S' and D' of Fig. 1), and the inversion charge is usually taken as $Q_i = C_{ox}(V_{GS'} - V_{th} - V_{D'S'}/2)$, where $V_{GS'}$ is the gate-source voltage, V_{th} is

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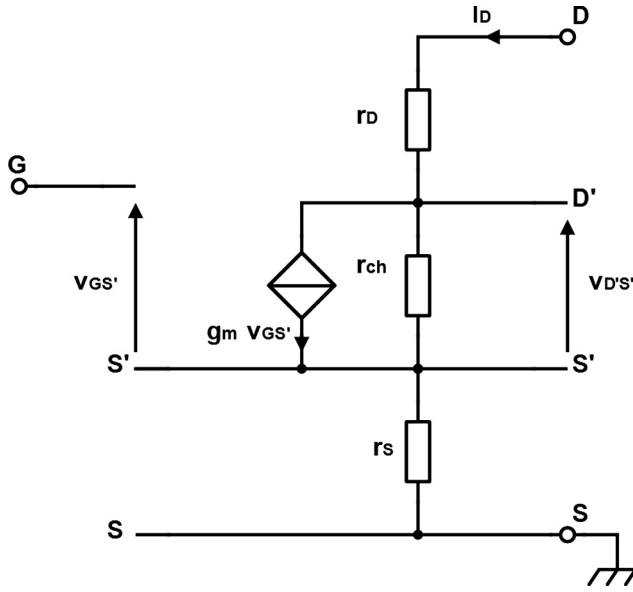


Fig. 1. MOSFET resistive model. The source (between nodes S and S'), the channel (between S' and D') and the drain (between D' and D) are the three resistive regions that compose the MOSFET.

the threshold voltage, and C_{ox} is the gate oxide capacitance per unit area.

The carrier mobility can be altered by several phenomena such as phonon scattering (thermal vibrations of the crystal lattice), Coulomb scattering (ionized impurities that deviates carriers) and surface roughness scattering. The mobility limited by the phonon scattering μ_{ph} , by the Coulomb scattering μ_c and by the surface roughness scattering μ_{sr} lead to the effective carrier mobility according to Matthiessen's rule [2]

$$\frac{1}{\mu_{eff}} = \frac{1}{\mu_{ph}} + \frac{1}{\mu_c} + \frac{1}{\mu_{sr}} \quad (2)$$

In strong inversion (i.e. for strong transversal electric field) the Coulomb scattering is negligible, leading to $\mu_{eff}^{-1} \approx \mu_{ph}^{-1} + \mu_{sr}^{-1}$. The effective carrier mobility of the inversion channel carriers μ_{eff} depends of the low-field mobility μ_0 , as seen in [3]

$$\mu_{eff} = \frac{\mu_0}{1 + \theta_{10} \left(V_{GS'} - V_{th} - \frac{V_{DS'}}{2} \right) + \theta_2 \left(V_{GS'} - V_{th} - \frac{V_{DS'}}{2} \right)^2} \quad (3)$$

θ_{10} is called the intrinsic mobility attenuation factor and is due to phonon scattering, while θ_2 is the second order mobility attenuation factor, which is due to surface roughness scattering. Assuming that $R_S = R_D = R_{access}/2$, the drain current expression in the linear operation regime can be obtained by replacing Q_i and μ_{eff} in (1)

$$I_D = \frac{G_M V_{DS} (V_{GS} - V_{th} - \frac{V_{DS}}{2})}{1 + (\theta_{10} + G_M R_{access}) (V_{GS} - V_{th} - \frac{V_{DS}}{2}) + \theta_2 (V_{GS} - V_{th} - \frac{V_{DS}}{2})^2} \quad (4)$$

where $G_M = \mu_0 C_{ox} W / L$ is the transconductance parameter. The extrinsic mobility attenuation factor is defined by $\theta_1 = \theta_{10} + G_M R_{access}$ and takes into account the drain current attenuation due to the phonon scattering and the access resistance. The transistor transconductance is defined as the variation of the drain current with respect to the gate voltage $g_m = \partial I_D / \partial V_{GS}$ and yields

$$g_m = G_M V_{DS} \frac{1 - \theta_2 (V_{GS} - V_{th} - \frac{V_{DS}}{2})^2}{\left[1 + \theta_1 (V_{GS} - V_{th} - \frac{V_{DS}}{2}) + \theta_2 (V_{GS} - V_{th} - \frac{V_{DS}}{2})^2 \right]^2} \quad (5)$$

2.2. Y function method

The Y function has two main advantages: it does not depend on the extrinsic mobility attenuation factor θ_1 and thus is independent on the access resistance R_{access} , and does not use the drain current second derivative, in opposition to some other methods. However, it assumes that the low-field mobility μ_0 is independent of the gate length L . The Y function is based on the drain current and transconductance equations as [3–5]

$$Y(V_{GS}) = \frac{I_D}{\sqrt{g_m}} = \sqrt{\frac{G_M V_{DS}}{1 - \theta_2 (V_{GS} - V_{th} - \frac{V_{DS}}{2})^2}} \left(V_{GS} - V_{th} - \frac{V_{DS}}{2} \right) \quad (6)$$

In moderate inversion, when $\theta_2 (V_{GS} - V_{th} - \frac{V_{DS}}{2})^2 \ll 1$, $Y(V_{GS})$ should be linear and a linear fitting gives access to G_M and V_{th} from the slope and the x-intercept, respectively. These estimations allow building the effective mobility attenuation factor θ_{eff} and thus determining the values of θ_1 and θ_2 from

$$\theta_{eff}(V_{GS}) = \frac{G_M V_{DS}}{I_D} - \frac{1}{V_{GS} - V_{th} - \frac{V_{DS}}{2}} = \theta_1 + \theta_2 \left(V_{GS} - V_{th} - \frac{V_{DS}}{2} \right) \quad (7)$$

This function should be linear and θ_1 and θ_2 can be extracted from the y-intercept and the slope, respectively. The following step consists in extracting the access resistance R_{access} and the intrinsic mobility attenuation factor θ_{10} from the θ_1 and G_M values linear plot, as

$$\theta_1 = \theta_{10} + G_M R_{access} \quad (8)$$

Once the parameters G_M , V_{th} , θ_1 and θ_2 have been extracted, another iteration of the Y_n function method can be performed with a new Y_{n+1} function as written in

$$Y_{n+1}(V_{GS}) = Y_n(V_{GS}) \sqrt{1 - \theta_2 \left(V_{GS} - V_{th} - \frac{V_{DS}}{2} \right)^2} \quad (9)$$

This should eliminate the influence of θ_2 in (5) and new values of G_M , V_{th} , θ_1 and θ_2 can be extracted, until the values converge. Examples of drain current and transconductance concordance between the experimental data and models of the drain current and of the transconductance described by Eqs. (4) and (5) are shown in Fig. 2. The right y-axis indicates that the absolute relative error is less than 0.1% for the drain, which reflects the good precision of this extraction procedure.

Several parameters can be extracted once the Y function method has been applied to devices with different gate dimensions [3]. The effective gate dimensions L and W can be determined using G_M and the mask gate dimensions L_m and W_m , as

$$G_M = \frac{\mu_0 C_{ox} W}{L} = \frac{\mu_0 C_{ox}}{L} (W_m - \Delta W) \quad (10.a)$$

$$\frac{1}{G_M} = \frac{L}{\mu_0 C_{ox} W} = \frac{1}{\mu_0 C_{ox} W} (L_m - \Delta L) \quad (10.b)$$

Assuming μ_0 is constant, both equations lead to a straight line, where the slope and the x-intercept of $G_M(W_m)$ or $G_M^{-1}(L_m)$ lead to μ_0 and ΔW or ΔL , respectively.

The extrinsic mobility attenuation factor θ_1 depends of the gate dimensions W and L , and on the access resistance R_{access} according to (7). Consequently a $\theta_1(G_M)$ function will give access to the intrinsic mobility attenuation factor θ_{10} and the drain-source resistance R_{access} .

The channel length reduction ΔL can also be determined with another method that consists in tracing the evolution of the total

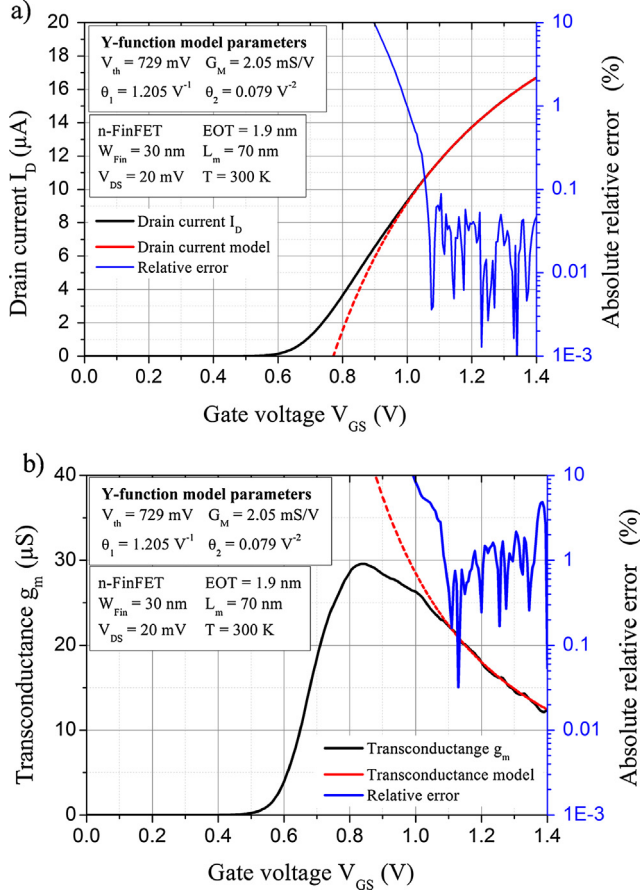


Fig. 2. MOSFET drain current (a) and transconductance (b). The Y function method leads to a model for both characteristics in strong inversion (in red). The relative error (in blue) in strong inversion remains below 5% for the transconductance. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

resistance (i.e. of I_D/V_{DS}) at different gate voltages V_{GS} for different gate mask lengths, using the channel resistivity ρ_{ch} in the relation

$$R_{tot} = \frac{L}{W} \rho_{ch} + R_{access} = \frac{L_m - \Delta L}{W} \rho_{ch} + R_{access} \quad (11)$$

All lines should intercept at the same point, which coordinates are ΔL and R_{access} , as seen in Fig. 3.

3. Low frequency noise

3.1. Low frequency noise in MOSFETs

The characterization of the low frequency noise in field-effect transistors is performed from measurements of the noise power spectral density (PSD), which is usually referred either to the drain current (S_{ID} in A 2 /Hz) or to the gate voltage (S_{VG} in V 2 /Hz). Both quantities are linked together by the MOSFET transconductance as $S_{VG} \cdot g_m^2 = S_{ID}$.

The noise mainly originates from the resistive parts of the transistors, which are the source region (r_s), the drain region (r_D) and the channel (r_{ch}). The low frequency noise equivalent circuit is given in Fig. 4, where the resistive (noisy) parts have been modeled by uncorrelated current noise sources. The drain region is between the points D and D' , the channel is between D' and S' and the source region is between S' and S . From this schematic we can write the total drain current noise $i_{D,tot}$ as

$$i_{D,tot} = g_m \cdot v_{GS'} + g_{ch} \cdot v_{D'S'} + i_{ch} \quad (12)$$

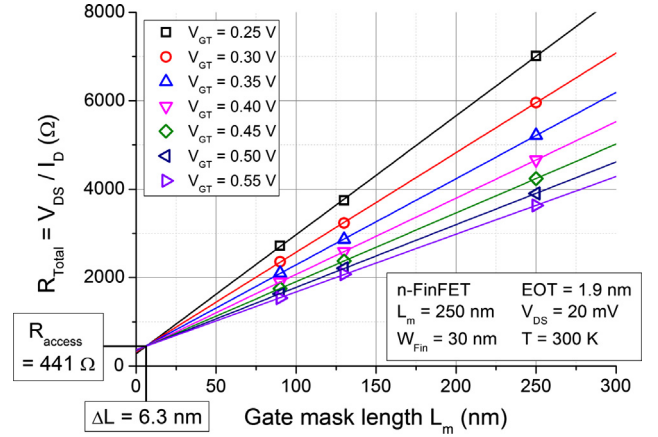


Fig. 3. MOSFET static parameters extraction. The access resistance R_{access} and the channel length reduction ΔL are obtained from the intercept of $R_{tot}(L_m)$ lines at different V_{GT} .

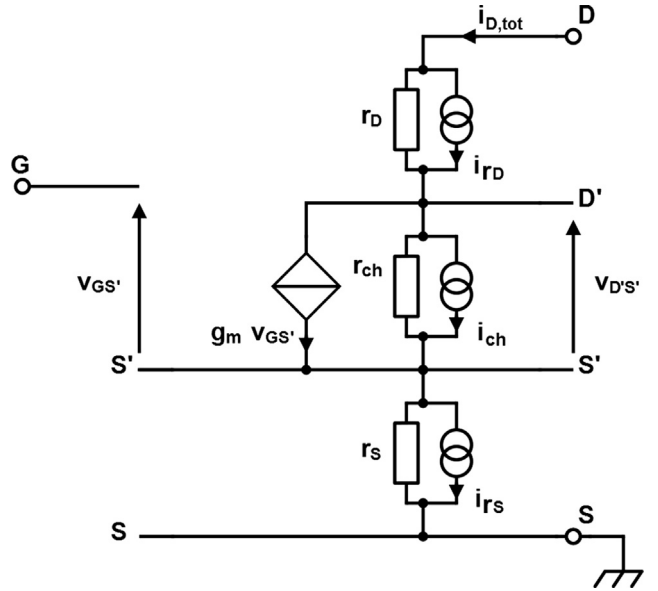


Fig. 4. MOSFET noise model with a representation of the three noisy region, which are the drain region (between nodes D and D'), the inversion channel (between nodes D' and S') and the source region (between nodes S' and S).

where $g_{ch} = r_{ch}^{-1}$ is the channel conductance. The $v_{GS'}$ and $v_{D'S'}$ voltages are

$$v_{GS'} = -v_{S'S} = -r_s(i_{D,tot} - i_{rs}) \quad (13.a)$$

$$v_{D'S'} = -v_{D'D} - v_{S'S} = -r_D(i_{D,tot} - i_{rD}) - r_s(i_{D,tot} - i_{rs}) \quad (13.b)$$

From (12) and (13) the total current noise in MOSFETs can be expressed by

$$i_{D,tot} = \frac{i_{ch} + i_{rD} \cdot g_{ch} r_D + i_{rs} \cdot r_s (g_m + g_{ch})}{1 + g_m r_s + g_{ch} (r_s + r_D)} \quad (14)$$

The total current noise PSD $S_{I_{D,tot}}$ can be written as a function of the channel current noise PSD $S_{I_{ch}}$, the drain region noise PSD $S_{I_{rD}}$ and the source region noise PSD $S_{I_{rs}}$

$$S_{I_{D,tot}} = \frac{S_{I_{ch}} + S_{I_{rD}} (g_{ch} r_D)^2 + S_{I_{rs}} [r_s (g_m + g_{ch})]^2}{[1 + g_m r_s + g_{ch} (r_s + r_D)]^2} \quad (15)$$

It is convenient to assume that the source and the drain region behave the same way, so that $r_D = r_S = r_{\text{access}}/2$ and $S_{I_{D,S}} = S_{I_{S,D}}$. In linear operation can be considered $g_m \ll g_{ch}$, then (15) yields

$$S_{I_{D,tot}} = S_{I_{ch}} \frac{(r_{tot} - r_{\text{access}})^2}{r_{tot}^2} + S_{I_R} \frac{r_{\text{access}}^2}{2r_{tot}^2} \quad (16)$$

where r_{tot} is the transistor total dynamic resistance (between the points D and S in Fig. 3).

3.2. White noise

The white noise is characterized by its level K_w , which is constant in the frequency domain, and has two main origins. The first one is the thermal noise (Johnson noise [6] or Nyquist noise [7]), which is due to the thermal motions of the electrons in the crystal lattice. Its voltage noise PSD is equal to $S_{V_G} = 4 k \cdot T \cdot R$, k being the Boltzmann constant, T the temperature and R the resistance of the piece of material.

The second one is the shot noise and has been described by Schottky in 1918 [8]. It is due to the random number of charge carriers that flows across a potential barrier at a given time. For an average current I flowing through a potential barrier, the current noise PSD is $S_{ID} = 2 q \cdot I$.

3.3. Generation-recombination noise

The generation-recombination (GR) noise is represented in the frequency domain by a Lorentzian shape, which consists in a plateau of level A followed by a $1/f^2$ decrease starting at the characteristic frequency f_0 . The latter parameter is linked to the characteristic time constant τ . GR noise is due to traps (in the gate oxide and in the depletion region) that randomly capture and release channel carriers, resulting in fluctuations of the available carriers for the current flow. Those traps may also induce changes in the carrier mobility, the scattering coefficient, the electric field and the potential barrier. It becomes significant when the Fermi energy level is close to the trap energy level, meaning that it is very sensitive to temperature [9].

Several Lorentzians can appear in the frequency spectrum, each having different values of A_i and $f_{0,i}$ (and τ_i) following (17) [10]. If the trap characteristic time constants are distributed in a precise way, the sum of the Lorentzians can have a $1/f$ shape over several decades of frequency.

$$S_{V_{G,lor_i}} = \frac{A_i}{1 + \left(\frac{f}{f_{0,i}}\right)^2} = \frac{A_i}{1 + (2\pi f \tau_i)^2} \quad (17)$$

If the gate surface is small enough (usually below $1 \mu\text{m}^2$) only a few traps are involved in the generation-recombination noise, meaning that the number of channel carriers being captured and released at a given time is also small. Thus the drain current displays a typical behavior in the time domain as a burst noise (also called Random-Telegraph-Signal, or RTS noise) appears superposed on the “classical” noise.

From [11] the trap characteristic time constant τ of a generation-recombination process is given by

$$\tau = \frac{1}{C_n(n_1 + n) + C_p(p_1 + p)} \quad (18)$$

with C_n and C_p the electrons and holes capture coefficients, respectively, n and p the free electrons and holes concentrations, respectively, n_1 and p_1 the concentrations of free electrons and holes under when the Fermi level coincides with the noisy trap level. In any case n , p , n_1 and p_1 vary with the temperature following an exponential dependency, which means that Lorentzians character-

istic time constants are temperature dependent. Nevertheless the variation of the characteristic time constants on the gate voltage depends on the localization of the traps in the transistor (either in the depletion film or at the Si-SiO₂ interface).

For Lorentzians that are related to interface traps, the Fermi level always coincides with the traps energy level [12] and $n = n_1$ and $p = p_1$, then $\tau = [2(C_n \cdot n_1 + C_p \cdot p_1)]^{-1}$. As the gate voltage $|V_{GS}|$ increases, the distance between the Fermi level and the edge of the majority carrier band increases, which leads to an increase of τ . However for a further increase of $|V_{GS}|$ the Fermi level approaches the minority carrier band and τ then decreases.

In the depletion region, free electrons and holes are practically absent, so the approximation $n = p = 0$ can be used in (18) and yields $\tau = (C_n \cdot n_1 + C_p \cdot p_1)^{-1}$. Traps close to the midgap have a concentration of $n_1 \approx p_1 \approx n_i$, where n_i is the intrinsic carrier concentration. This leads to a value of $\tau = [n_i \cdot (C_n + C_p)]^{-1}$, which is independent of the gate voltage $|V_{GS}|$.

3.4. 1/f noise

Several models have been proposed in order to explain $1/f$ noise in the inversion layer. The carrier number fluctuations model (ΔN) was first described by McWhorter in 1957 and it suggests that carrier tunneling into the oxide is the origin of the $1/f$ noise [13]. Carrier mobility fluctuations model ($\Delta\mu$) described by Hooge in 1969 explains that the $1/f$ noise is due to phonon scattering [14]. A carrier number fluctuations correlated to mobility fluctuations model ($\Delta N + \Delta\mu$) has been suggested in 1990 [15,16]. It states that oxide traps that capture and release channel carriers change the number of available carriers, which leads to surface mobility fluctuations.

In the case of carrier mobility fluctuations model, the gate voltage noise PSD of the channel can be written as [3]

$$S_{V_G} = \frac{q\alpha_H}{fWLC_{ox}} (V_{GS} - V_{th}) [1 + \theta_1 (V_{GS} - V_{th})]^2 \quad (19)$$

where α_H is the Hooge parameter and depends of the crystal quality. In weak inversion, the normalized drain current noise PSD S_{I_D}/I_D^2 is inversely proportional to $V_{GS} - V_{th}$.

For the $\Delta N + \Delta\mu$ model, the gate voltage noise PSD can be expressed by [17]

$$S_{V_G} = S_{V_{fb}} (1 \pm \alpha_C \mu_{eff} C_{ox} V_{GT})^2 \quad (20)$$

where $S_{V_{fb}}$ is the flat-band spectral density and α_C is the Coulomb scattering coefficient. The + and the − signs apply for donor and acceptor traps, respectively. This equation requires no assumption about the mechanism behind the fluctuations in the flat-band voltage [18]. However if $\alpha_C \mu_{eff} C_{ox} V_{GT} \ll 1$, then $S_{V_G} = S_{V_{fb}}$ and only carrier number fluctuations noise is observed.

As the noise of the access resistance can be modeled by the formula $S_{I_D, \text{access}} = \frac{K_r}{f} I_D^2$ [19], with K_r being the access resistance noise parameter, we can write from (16), (19) and (20) the total gate voltage spectral density in ohmic region operation

$$S_{V_G} = \frac{(r_{tot} - r_{\text{access}})^2}{r_{tot}^2} \frac{q\alpha_H}{fWLC_{ox}} (V_{GS} - V_{th}) [1 + \theta_1 (V_{GS} - V_{th})]^2 + \frac{K_r}{f} \frac{r_{\text{access}}^2}{2r_{tot}^2} \frac{I_D^2}{g_m^2} \quad (21.a)$$

$$S_{V_G} = \frac{(r_{tot} - r_{\text{access}})^2}{r_{tot}^2} S_{V_{fb}} (1 + \alpha_C \mu_{eff} C_{ox} V_{GT})^2 + \frac{K_r}{f} \frac{r_{\text{access}}^2}{2r_{tot}^2} \frac{I_D^2}{g_m^2} \quad (21.b)$$

The first term of (21.a) and (21.b) corresponds to the $\Delta\mu$ model and the $\Delta N + \Delta\mu$ model, respectively. The second term corresponds to the access resistance noise, usually observed in strong inversion.

4. Methodology

4.1. Experimental bench

The device under test (DUT) is placed into a Lakeshore TTP4 cryogenic probe station. The low frequency noise experimental bench uses home-made electronics and a HP 3562A dynamic signal analyzer. A simplified schematic of these electronics is shown in Fig. 5. The transistor is biased with V_{DS} and V_{GS} voltage sources, the resulting drain current I_D flows through the retroaction resistance of the transimpedance amplifier and the voltage measured by the voltmeter allows checking the bias point, as $V_{OUT} = V_{DS} + R \cdot I_D$.

The drain noise current i_D is converted into a voltage after the transimpedance stage, then this voltage passes through the high-pass filter (cut-off frequency of 0.08 Hz) and is amplified by an instrumentation amplifier. The spectrum analyzer finally provides the output voltage noise PSD $S_{V_{out}}$ from 1 Hz to 100 kHz.

In order to get the input-referred noise PSD (i.e. the gate voltage noise PSD), white noise is injected into the gate (v_{gs} voltage source in Fig. 5) after the output PSD measurements. The ratio between the output voltage and the white noise voltage gives the whole system gain, which allows to get the gate voltage noise PSD as $S_{V_G} = S_{V_{out}} / \text{gain}^2$ and to get rid of the system bandwidth limitation.

4.2. Noise model

The low frequency noise is generally due to the contributions of three distinct noises, which are the white noise (of a constant level K_w), the $1/f$ noise (of level K_f) and the generation-recombination noise. The latter can present several Lorentzians (of plateau level A_i and characteristic frequency $f_{0,i}$) in the noise PSD. The gate voltage noise PSD S_{V_G} can be seen as the sum of these three noise sources and is written

$$S_{V_G}(f) = K_w + \frac{K_f}{f^\gamma} + \sum_{i=1}^N \frac{A_i}{1 + (f/f_{0,i})^2} \quad (22)$$

A value of $\gamma = 1$ means that the traps are uniformly distributed in the gate oxide depth. Fig. 6 shows a gate voltage noise PSD model and its different components using (11).

However, in order to better extract the noise parameters, one can use a noise PSD normalized by the frequency, as displayed in Fig. 7. The $1/f$ noise is then represented by a constant value while the Lorentzians correspond to bumps centered around their characteristic frequency $f_{0,i}$. As it is more convenient to find $f_{0,i}$, it reduces the estimation errors on the plateau levels A_i . Consequently using the model described by (22) leads to the white noise level K_w , the $1/f$ noise level K_f and the Lorentzians parameters A_i and $f_{0,i}$.

4.3. $1/f$ noise mechanism and oxide trap density

In order to determine the $1/f$ noise mechanism, one has to perform noise measurements as a function of the gate voltage

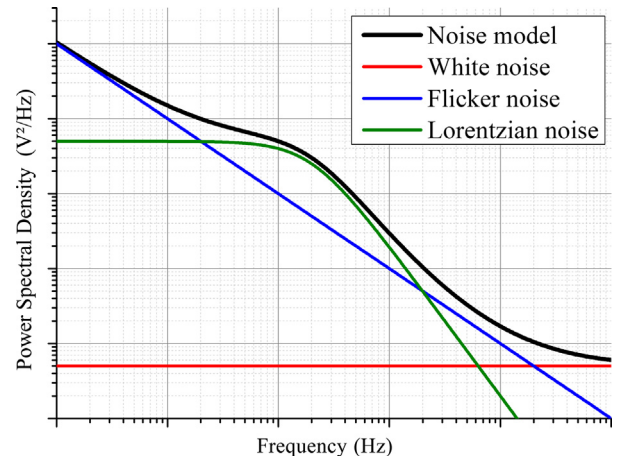


Fig. 6. Total noise PSD model (in black). The total noise PSD is the sum of the white noise, the $1/f$ noise and one Lorentzian.

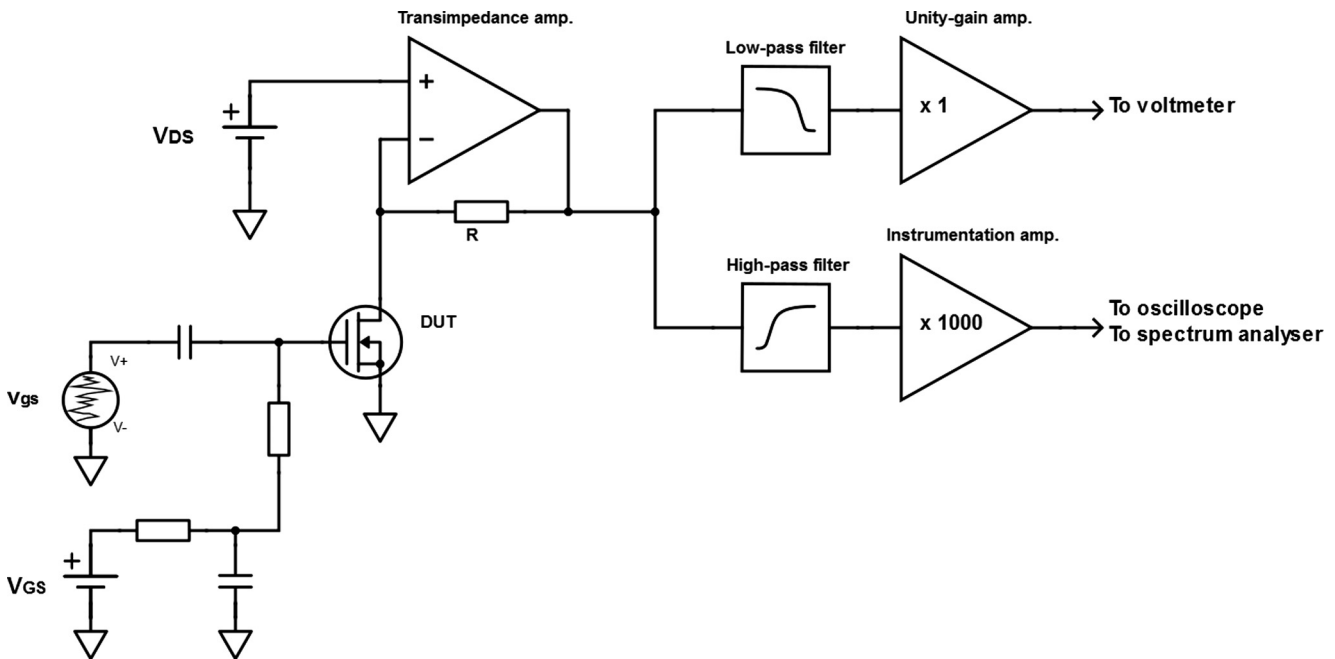


Fig. 5. Simplified schematic of the low frequency noise measurements bench.

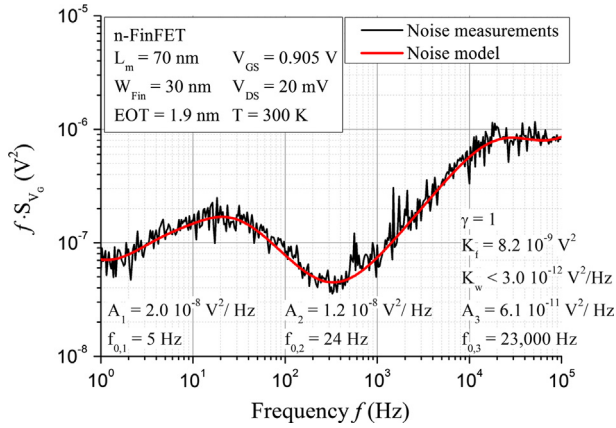


Fig. 7. Frequency-normalized gate voltage noise PSD. The noise model (20) allowed to extract the noise parameters (white noise level K_w , $1/f$ noise level K_f and characterization of three Lorentzians).

V_{GS} (or drain current I_D). For each noise PSD, a noise model must be performed using (21), which leads to the $1/f$ noise level K_f . The evolution of K_f as a function of V_{GS} (or I_D) leads to the $1/f$ noise mechanism, according to (21.a) and (21.b). Table 1 summarizes the evolution of S_{V_G} and S_{I_D}/I_D^2 using (21.a) and (21.b).

As the values of the effective mobility μ_{eff} , the mobility attenuation factor θ_1 and the access resistance r_{access} have been extracted with static measurements, the coulomb scattering coefficient α_C and the flat-band voltage PSD $S_{V_{fb}}$ (for $\Delta N + \Delta\mu$ model) or the Hooge parameter α_H (for $\Delta\mu$ model) can be determined. In the case of number fluctuations correlated to mobility fluctuations, the flat-band voltage PSD leads to an estimation of the oxide trap density N_T , according to the formula

$$S_{V_{fb}} = \frac{q^2 k T \lambda N_T}{f^2 W L C_{ox}} \quad (23)$$

where λ is the tunneling attenuation length in the gate oxide (≈ 1 Å for Si/SiO₂).

4.4. Silicon film traps: noise spectroscopy

As for studying the $1/f$ noise, the Lorentzian parameters (plateau level A_i and characteristic frequency $f_{0,i}$) are obtained from noise PSD modelling at different values of V_{GS} and at fixed temperature, using (22). As explained above, Lorentzians with a characteristic frequency that is independent of the gate voltage are considered to be related to traps in the depletion region whereas Lorentzians with a characteristic frequency that depends on V_{GS} are located in the gate oxide. However their characteristic frequency should vary with the temperature in both cases. Lorentzians with fixed $f_{0,i}$ as a function of V_{GS} have been studied in order to identify the physical nature of the Si film traps. To do so, one must perform noise spectroscopy, i.e.

noise measurements at different temperatures and at constant drain current I_D .

The characteristic time constant τ_i of a single trap varies with temperature, according to the expression [1]

$$\ln(\tau_i T^2) = (E_C - E_T) \frac{q}{kT} + \ln \left(\frac{h^3}{4k^2 \sigma_n \sqrt{6\pi^2 M_c m_e^{*1/2} m_h^{*3/2}}} \right) \quad (24)$$

where σ_n is the trap capture cross section and $E_C - E_T$ is the difference between the conduction band energy level and the trap energy level. k is the Boltzmann constant, h is the Planck constant, m_e^* and m_h^* are the effective mass of electrons and holes, respectively, and M_c is the number of conduction band energy minima.

From the characteristic time constant τ_i and the temperature, one can trace the evolution of $\ln(\tau_i T^2)$ with respect to q/kT , which is an Arrhenius diagram and should give a straight line. By using (23) the estimated capture cross section σ_n and energy difference $E_C - E_T$ can be determined and compared with data in the literature. Indeed Arrhenius diagrams of common traps are known from other methods, like Deep Level Transient Spectroscopy. If experimental points fit one particular line, then the associated Lorentzians correspond to the same particular trap.

Once traps have been identified, the associated plateau levels A_i should be traced against the characteristic time constants τ_i . This should give a straight line and lead to the surface (effective) trap density N_{eff} for each trap, according to

$$A_i = \frac{q^2 N_{eff} \tau_i}{W L C_{ox}^2} = \frac{q^2 W_d N_T \tau_i}{W L C_{ox}^2 B} \quad (25)$$

where q is the elementary charge, N_T is the volume trap density, W_d is the depletion depth and B is a coefficient. B is usually predicted to be 1/3 for planar devices [19], however this does not apply well for multi-gate devices [20]. Consequently, even though silicon film traps are related to a volume phenomenon, the surface trap density N_{eff} will be extracted as it requires no assumption of the coefficient B .

5. Conclusion

This paper has presented the main noise parameters that can be extracted, either from the $1/f$ noise or the generation-recombination noise. First static characteristics need to be performed in order to extract DC parameters that will be used for noise characterization, such as the threshold voltage V_{th} , the low-field mobility μ_0 , the mobility attenuation factor θ and the access resistance r_{access} . Then gate voltage noise measurements can be performed in order to determine the $1/f$ noise mechanism and the gate oxide trap density N_T . Finally a low frequency noise spectroscopy leads to the identification of depletion region traps and an estimation of their effective densities N_{eff} .

Table 1

Evolution of the $1/f$ noise PSD (S_{V_G} and S_{I_D}) with respect to the gate overdrive voltage V_{GT} or the drain current I_D , according to the McWhorter model ($\Delta N + \Delta\mu$), the Hooge model ($\Delta\mu$), and the access resistance noise model.

	Channel noise: $\Delta N + \Delta\mu$ fluctuations		Channel noise: $\Delta\mu$ fluctuations	Access resist. noise
	$\alpha_C \mu_{eff} C_{ox} V_{GT} \ll 1$	$\alpha_C \mu_{eff} C_{ox} V_{GT} \gg 1$		
$S_{V_G} \propto$	Constant	$(1 \pm \alpha_C \mu_{eff} C_{ox} V_{GT})^2$	$V_{GT} [1 + \theta_1 V_{GT}]^2$	$\frac{I_D^2}{(r_{tot} g_m)^2}$
$\frac{S_{I_D}}{I_D} \propto$	$\left(\frac{g_m}{I_D}\right)^2$	$\left(\frac{g_m}{I_D}\right)^2 [1 \pm \alpha_C \mu_{eff} C_{ox} V_{GT}]^2$	$\frac{1}{V_{GS} - V_{th}}$	$\frac{1}{r_{tot}}$

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